What minimum number of ground stations is needed? Give the position equations.

Stations lie on one meridian at N and S (unit:

Both receive a satellite signal after emission. Find its maximum altitude

above ground For small latitude errors at , find the error in

For a small error in time , find the error in .

. In , take Find the maximum altitude

in km. If , how large is the error in ? If

, what error in ? Earth is a sphere of radius ;

, and ground gravitational acceleration

. Now use reference-satellite motion to track an object. Neglect relativity. Each continuously emitted radio signal states its emission time and the satellite's position then. A receiver aboard the object, with no clock, obtains these data. To infer its motion state, its position and time on receiving a signal, what minimum number of different satellites must it receive simultaneously? Write the equations for the object's position and time at reception.

. Special relativity slows moving clocks; general relativity slows clocks in a gravitational field. Include both effects. Satellite and station clocks have aligned zero readings. Orbit height ; radius , light speed , and ground acceleration are as given in part .

By special relativity, estimate the reading difference after on Earth: ground minus satellite. Use uniform rectilinear time dilation for circular motion.

The GR slowing factor is where is the magnitude of gravitational potential: potential energy per unit mass, zero at infinity. After on the ground clock, find the satellite-minus-ground clock reading.

3.

A refrigerator sends heat from a cold reservoir, plus energy from external work, to a hot reservoir, cooling the cold one and warming the hot one. If it works between absolute temperatures (hot) and (cold), taking heat